Gated Cross-Attention for Explainable Multimodal Cancer Diagnosis

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***Abstract*—**Clinical oncology depends on the joint interpretation of histopathology, patient narratives, and longitudinal laboratory data. Existing automated systems usually process these modalities in separate pipelines, which limits their ability to reflect the evidence used in multidisciplinary decision making. This paper presents the Explainable Hybrid AI System for Multimodal Cancer Diagnosis and Treatment Recommendation (EHASMCDTR), a tri-modal architecture that combines a ResNet-50 vision encoder, a BioBERT language encoder, and a multilayer perceptron (MLP) tabular encoder. Gated cross-attention aligns the resulting representations, which feed a multi-task prediction head for tumor classification, TNM staging, therapeutic ranking, and survival estimation. Homoscedastic uncertainty weighting balances the task losses, while Grad-CAM and SHAP provide visual, textual, and tabular attributions. On a four-class cohort of 4,500 records (breast, lung, and colorectal cancer, and normal/benign tissue), split 70/30 into 3,150 training and 1,350 test samples and trained for 40 epochs, EHASMCDTR obtained 97.17% aggregate classification accuracy and a macro/weighted F1-score of 0.9717. Class-wise accuracies were 97.80% for breast carcinoma, 97.20% for colorectal carcinoma, and 96.50% for lung carcinoma. These results indicate that gated multimodal fusion can combine complementary evidence while retaining an auditable explanation pathway.

***Index Terms*—**Biomedical text mining, computational pathology, deep learning, explainable artificial intelligence, multimodal fusion, precision oncology.

I. INTRODUCTION

Cancer diagnosis and treatment planning require evidence from tissue morphology, clinical history, and physiological measurements. In practice, these sources are spread across pathology systems and electronic health records, and limited specialist availability and inter-observer variability can delay interpretation. Deep learning has advanced isolated tasks such as image-based screening, clinical text extraction, and survival modeling, but a model that observes a single channel cannot reproduce the evidence integration performed during tumor-board review.

EHASMCDTR addresses this limitation by processing digitized histopathology images, unstructured clinical narratives, and structured laboratory records within a single architecture. Modality-specific encoders produce intermediate representations that are exchanged through gated cross-attention before multi-task prediction. The design has four objectives: to unify three evidence streams, to reduce modality dominance during fusion, to jointly predict diagnosis, stage, therapy ranking, and survival, and to produce an explanation trail that links outputs to image regions and nonvisual features.

The principal contributions are: 1) a tri-modal oncology framework; 2) gated cross-attention for selective cross-modal interaction; 3) uncertainty-weighted multi-task optimization; and 4) native Grad-CAM and SHAP attribution. The remainder of this paper is organized as follows: Section II reviews related work, Section III formulates the problem, Section IV describes the methodology, Section V presents the experimental setup, Section VI reports and discusses the results, and Sections VII and VIII address limitations and conclude.

II. RELATED WORK

A. Medical Image Analysis

Convolutional neural networks have achieved strong results in clinical image analysis. Large-scale skin-lesion classification demonstrated the feasibility of dermatologist-level image prediction [1], and CheXNet showed the value of dense convolutional representations for chest radiograph interpretation [2]. In digital histology, patch-based analysis captures nuclear morphology and tissue architecture, but image-only models cannot use patient history or laboratory measurements. ResNet-50 is therefore adopted here as a

practical visual backbone for extracting hierarchical tissue features.

B. Clinical Language Modeling

Pathology reports and clinical notes record margins, grade, nodal involvement, staging, and treatment history. Biomedical transformer models such as BioBERT adapt contextual language representations to scientific and clinical terminology [3]. Such models can identify entities and relations that general-domain language models recover poorly, but text-only systems omit the spatial information contained in tissue images.

C. Multimodal Fusion

Multimodal systems may combine raw features, decisions, or intermediate representations. Intermediate fusion allows interaction between modalities, but a high-capacity stream can dominate optimization and suppress weaker, complementary signals. Gated cross-attention addresses this by learning how much information from one modality should influence another. EHASMCDTR applies the mechanism to image, text, and tabular embeddings instead of relying on direct concatenation.

D. Explainability in Clinical AI

Clinical deployment requires explanations that let users inspect model reasoning. Grad-CAM produces spatial attribution maps from the gradients of a convolutional network [4], whereas SHAP assigns additive feature contributions based on Shapley values [5]. Grad-CAM is spatially informative but relatively coarse; SHAP suits tabular variables and tokens. The proposed system reports both forms of attribution within a single inference workflow. Table I contrasts these modality and explanation strategies with the corresponding EHASMCDTR design choices.

TABLE I. Comparison of Modality and Explanation Strategies

Approach	**Modalities**	**Limitation**	**EHASMCDTR response**
CNN baseline	Image	Ignores clinical context	Adds narrative and laboratory streams
BioBERT/NLP	Text	Ignores tissue topology and physiology	Fuses text with image and tabular embeddings
Basic multimodal fusion	Image + text + data	Modality dominance and negative transfer	Gated cross-attention and uncertainty weighting
Post-hoc explanations	Variable	Disconnected visual and feature explanations	Joint Grad-CAM and SHAP audit trail

III. PROBLEM FORMULATION

Let a patient record be represented by the multimodal set $D = \{I, T, R\}$, where I denotes histopathology images, T denotes clinical narratives, and R denotes structured physiological and laboratory records. The model learns the mapping in (1).

$$f(I, T, R) \rightarrow (\hat{y}type, \hat{y}stage, \hat{y}tx, \hat{y}surv, \Phi). \quad (1)$$

In (1), ŷtype denotes the four-class tumor-type prediction (breast, lung, colorectal, or normal/benign), ŷstage the TNM stage, ŷtx a ranked therapeutic output, ŷsurv the estimated longitudinal survival probability, and Φ the explanation matrix of visual and feature-level attributions. The model is intended as clinical decision support, and its outputs require review by qualified professionals.

IV. PROPOSED METHODOLOGY

A. Data Acquisition and Preprocessing

The evaluation uses a four-class corpus covering breast, lung, and colorectal cancer and normal/benign tissue, with histopathology images and the corresponding clinical text and structured-record modalities. Histopathology image tiles are resized to 224 × 224 pixels and processed with Macenko stain normalization to reduce variation caused by staining and scanning protocols [6]. Random horizontal flips, affine rotations, and chromatic jitter are applied during training. Clinical notes are scrubbed of protected health information, segmented into sentences, and tokenized with a WordPiece vocabulary. Missing laboratory values are median-imputed, categorical variables are one-hot encoded, and continuous measurements are min–max scaled to [0, 1].

B. Modality-Specific Encoders

A ResNet-50 encoder extracts hierarchical visual features from tissue tiles, and its global-average-pooling output is denoted by vI. A BioBERT encoder maps the clinical narrative to contextual token representations, and the final [CLS] representation is used as vT. An MLP with fully connected layers, batch normalization, and ReLU activations processes the structured demographic and laboratory values and yields vR. Fig. 1 shows the end-to-end architecture.

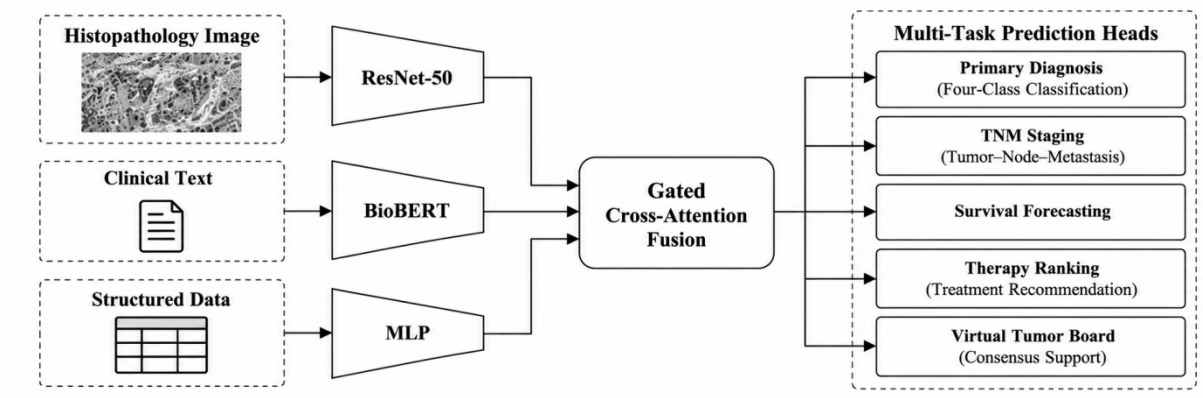


Fig. 1. End-to-end architecture of the EHASMCDTR multimodal framework.

C. Gated Cross-Attention Fusion

In cross-attention, a primary representation X supplies the query and a secondary representation Y supplies the key and value, as defined by the scaled dot-product operation in (2).

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V. \quad (2)$$

A learned sigmoid gate filters the attended output so that noisy or weakly aligned information is attenuated. The generic fusion update in (3) combines the gated output with a feedforward block.

$$\text{GCA}(X, Y) = \text{LayerNorm}(X + \text{FFN}(X + g \odot \text{Attention}(X, Y))). \quad (3)$$

where g is the learned gate, ⊙ denotes element-wise multiplication, and FFN is a feedforward network. Cross-modal interactions are applied before the fused patient-level representation is passed to the prediction heads.

D. Multi-Task Prediction and Loss

The shared representation feeds parallel heads for tumor type, TNM stage, therapeutic ranking, and survival probability. To account for differing task scales and noise levels, each task is assigned a learnable variance parameter σ_i^2, following homoscedastic uncertainty weighting [7]. The combined objective is given in (4).

$$L_{total} = \Sigma_i \left[(1 / 2\sigma_i^2) L_i + \log \sigma_i \right]. \quad (4)$$

The weighting in (4) reduces the influence of noisier tasks, while the log σ_i term discourages unbounded uncertainty. Classification uses cross-entropy, whereas survival prediction is specified with a Cox proportional-hazards loss in the original design.

E. Native Interpretability

Grad-CAM is computed at the final residual block of ResNet-50. For class c, the gradients of the class score y^c with respect to activation map A^k are globally averaged to obtain α_k^c, and the heatmap is then computed as in (5).

$$L\text{Grad-CAM}^c = \text{ReLU}(\Sigma_k \alpha_k^c A^k). \quad (5)$$

SHAP values are computed for structured biomarkers and relevant clinical tokens. Together with the Grad-CAM map in (5), they provide spatial and feature-level evidence for inspection by a medical team.

V. EXPERIMENTAL SETUP

A. Dataset and Partitioning

The experiments use four classes: breast cancer, lung cancer, colorectal cancer, and normal/benign tissue. The initial tile corpus contains 43,717 tissue-slide tiles: 32,199 colorectal, 7,909 breast, and 3,609 lung samples, 3,110 normal/benign samples. MD5 checksums were used to detect duplicate files before partitioning. Stratified sampling produced a cohort of 4,500 records, which was divided 70/30 into 3,150 training samples and 1,350 test samples. Images were represented as three-channel RGB arrays and normalized with ImageNet channel statistics (means [0.485, 0.456, 0.406]; standard deviations [0.229, 0.224, 0.225]).

B. Training Configuration

The implementation uses Python, PyTorch, Hugging Face Transformers, OpenCV, Pillow, SHAP, and Grad-CAM. The network is trained for 40 epochs with AdamW and weight decay. A cosine-annealing schedule with warm restarts varies the learning rate from 0.001 to 0.0003. An early-stopping criterion of seven consecutive validation epochs without improvement is also defined. ResNet-50, BioBERT, and the MLP are first initialized on their respective unimodal tasks, and the complete network is then fine-tuned end-to-end.

VI. RESULTS AND DISCUSSION

A. Classification Performance

On the 1,350-sample test set, EHASMCDTR achieved 97.17% aggregate accuracy and a macro/weighted F1-score of 0.9717. Table II reports the class-wise results, and Fig. 2 shows the normalized four-class confusion matrix. In the matrix, rows give the ground-truth class and columns the predicted class. The diagonal entries are 0.97 for breast, 0.96 for lung, 0.97 for colon, and 0.99 for normal/benign tissue, and no off-diagonal entry exceeds 0.02. The largest confusions are lung predicted as breast and colon predicted as lung (0.02 each). The macro and weighted F1-scores are identical in the reported results.

TABLE II. Primary Tumor Classification Performance

Malignancy type	**Accuracy (%)**	**F1-score**
Breast carcinoma	97.80	0.978
Colorectal carcinoma	97.20	0.972
Lung carcinoma	96.50	0.965
Normal/Benign	97.18	0.972
Aggregate (macro/weighted)	97.17	0.9717

Breast carcinoma produced the highest accuracy, followed by colorectal carcinoma. Lung classification was slightly lower, a difference we associate with greater cellular pleomorphism and overlapping pulmonary morphology. The results support the value of combining image features with clinical and structured representations, although a complete numerical ablation against each unimodal baseline is not included.

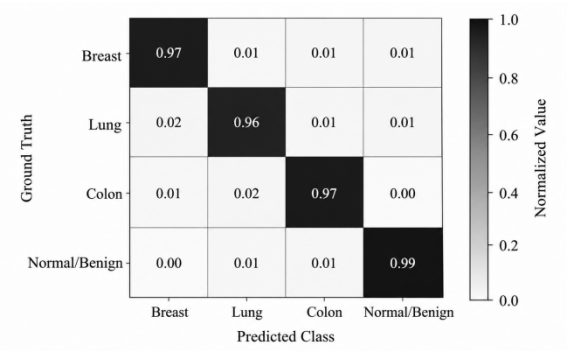


Fig. 2. Normalized confusion matrix for four-class cancer classification on the held-out test set. Rows: ground truth; columns: predicted class.

B. Fusion and Loss Weighting

The gating mechanism in (3) is intended to reduce dominance by the highest-dimensional modality, particularly text, by learning the proportion of cross-attended information passed to the shared representation. The uncertainty weighting in (4) addresses the differing gradient scales of the classification, staging, treatment-ranking, and survival objectives. Together, these mechanisms form a coherent optimization strategy, but their individual contributions should be quantified in future ablation experiments.

C. Interpretability and Clinical Use

Grad-CAM highlights the image regions that contributed most to the ResNet-50 prediction. Fig. 3 shows the original H&E image with its hematoxylin and eosin stain-deconvolution channels and the corresponding Grad-CAM activation, so the highlighted regions can be compared with the underlying staining patterns. Such maps indicate where the model attended; they do not by themselves establish clinical correctness.

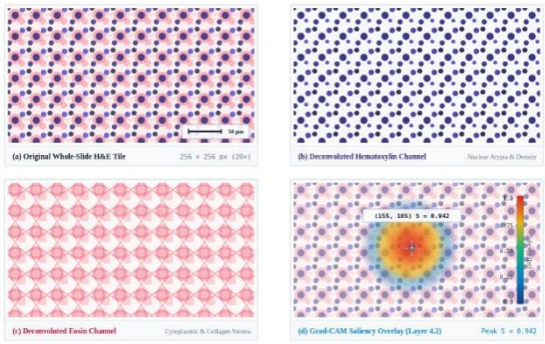


Fig. 3. Grad-CAM-based spatial localization with stain-deconvolution views for histopathological image interpretation.

SHAP provides feature-level attribution for the model output. Fig. 4 presents a SHAP waterfall plot in which each bar shows how a multimodal clinical or structured feature shifted the prediction for one case. These attributions are preliminary: reviewer agreement, calibration, confidence intervals, and external validation are not reported. EHASMCDTR should therefore be treated as decision support rather than an autonomous diagnostic or treatment system.

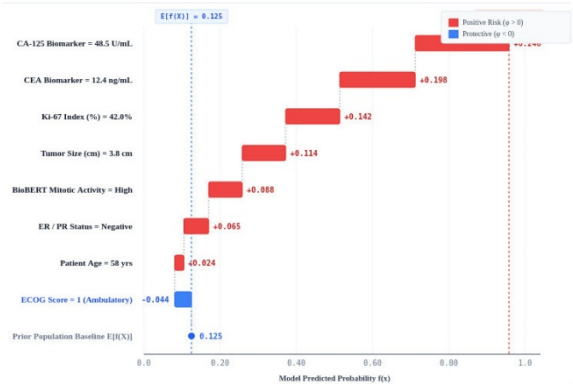


Fig. 4. SHAP-based feature attribution illustrating the contribution of multimodal clinical and structured features to the model prediction.

VII. LIMITATIONS AND FUTURE WORK

Several limitations remain. The evaluation cohort was obtained by subsampling a substantially imbalanced source corpus, which may reduce ecological validity. The results focus mainly on tumor classification; numerical performance for TNM staging, treatment ranking, survival prediction, calibration, and explanation faithfulness is not reported. The study also does not specify a separate patient-level split strategy for all modalities, an external test institution, confidence intervals, or prospective clinical evaluation. Future work should evaluate these endpoints, perform modality and component ablations, test robustness across scanners and hospitals, report subgroup performance, and assess explanation quality with blinded clinical reviewers.

VIII. CONCLUSION

This paper presented EHASMCDTR, a tri-modal framework that combines ResNet-50, BioBERT, and an MLP through gated cross-attention. The system jointly represents histopathology, clinical narratives, and laboratory records, and it produces multi-task outputs through uncertainty-weighted optimization and integrated Grad-CAM/SHAP explanations. On the four-class test set of 1,350 samples, it achieved 97.17% classification accuracy and a macro/weighted F1-score of 0.9717. The architecture is promising as an auditable research prototype, but broader validation and task-specific reporting are required before clinical deployment.

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